

KCET – Mathematics – 2019
VERSION CODE: B-2

Question 1: The inverse of the matrix $\begin{bmatrix} 2 & 5 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 3 \end{bmatrix}$

(A) $\begin{bmatrix} 3 & -15 & 5 \\ -1 & 6 & -2 \\ 1 & -5 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & -15 & 5 \\ -1 & 6 & -2 \\ 1 & -5 & -2 \end{bmatrix}$

(D) $\begin{bmatrix} 3 & -5 & 5 \\ -1 & -6 & -2 \\ 1 & -5 & 2 \end{bmatrix}$

Answer 1: (C)

Question 2: If P and Q are symmetric matrices of the same order then PQ-QP is

(A) Zero matrix

(B) Identity matrix

(C) Skew symmetric matrix

(D) Symmetric matrix

Answer 2: (C)

Question 3: If $3A + 4B' = \begin{bmatrix} 7 & -10 & 17 \\ 0 & 6 & 31 \end{bmatrix}$ and $2B - 3A' = \begin{bmatrix} -1 & 18 \\ 4 & 0 \\ -5 & -7 \end{bmatrix}$ then B =

(A) $\begin{bmatrix} -1 & -18 \\ 4 & -16 \\ -5 & -7 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 3 \\ -1 & 1 \\ 2 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 3 \\ -1 & 1 \\ 2 & -4 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & -3 \\ -1 & 1 \\ 2 & 4 \end{bmatrix}$

Answer 3: (B)

Question 4: If $A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$, Then $|ABB'| =$

- (A) 100
(B) 50
(C) 250
(D) -250

Answer 4: (D)

Question 5: If the value of a third order determinant is 16, then the value of the determinant formed by replacing each of its elements by its cofactor is

- (A) 256
(B) 96
(C) 16
(D) 48

Answer 5: (A)

Question 6: $\int x^3 \sin 3x dx =$

- (A) $-\frac{x^3 \cdot \cos 3x}{3} + \frac{x^2 \cdot \sin 3x}{3} + \frac{2x \cos 3x}{9} - \frac{2 \sin 3x}{27} + C$
(B) $-\frac{x^3 \cdot \cos 3x}{3} - \frac{x^2 \cdot \sin 3x}{3} + \frac{2x \cos 3x}{9} - \frac{2 \sin 3x}{27} + C$
(C) $-\frac{x^3 \cdot \cos 3x}{3} + \frac{x^2 \cdot \sin 3x}{3} - \frac{2x \cos 3x}{9} - \frac{2 \sin 3x}{27} + C$
(D) $\frac{x^3 \cdot \cos 3x}{3} + \frac{x^2 \cdot \sin 3x}{3} - \frac{2x \cos 3x}{9} - \frac{2 \sin 3x}{27} + C$

Answer 6: (A)

Question 7: The area of the region above X-axis included between the parabola $y^2 = x$ and the circle $x^2 + y^2 = 2x$ in square units is

(A) $\frac{2}{3} - \frac{\pi}{4}$

(B) $\frac{\pi}{4} - \frac{3}{2}$

(C) $\frac{\pi}{4} - \frac{2}{3}$

(D) $\frac{3}{2} - \frac{\pi}{4}$

Answer 7: (C)

Question 8: The area of the region bounded by Y-axis, $y = \cos x$ and $y = \sin x$ $0 \leq x \leq \frac{\pi}{2}$ is

(A) $\sqrt{2} + 1$ Sq. units

(B) $\sqrt{2} - 1$ Sq. units

(C) $2 - \sqrt{2}$ Sq. units

(D) $\sqrt{2}$ Sq. units

Answer 8: (B)

Question 9: The integrating factor of the differential equation $(2x + 3y^2) dy = y dx$ ($y > 0$) is

(A) $\frac{1}{x}$

(B) $\frac{1}{e^y}$

(C) $\frac{1}{y^2}$

(D) $\frac{1}{y^2}$

Answer 9: (C)

Question 10: The equation of the curve passing through the point (1, 1) such that the slope of the tangent at any point (x, y) is equal to the product of its co-ordinates is

(A) $2 \log y = x^2 - 1$

(B) $2 \log x = y^2 - 1$

(C) $2 \log x = y^2 + 1$

(D) $2 \log y = x^2 + 1$

Answer 10: (A)

Question 11: Foot of the perpendicular drawn from the point (1, 3, 4) to the plane $2x - y + z + 3 = 0$ is

- (A) (1, 2, -3)
- (B) (-1, 4, 3)
- (C) (-3, 5, 2)
- (D) (0, -4, -7)

Answer 11: (D)

Question 12: Acute angel between the line $\frac{x-5}{2} = \frac{y+4}{-1} = \frac{z+4}{1}$ the plane $3x - 4y - z + 5 = 0$ is

- (A) $\cos^{-1}\left(\frac{5}{2\sqrt{3}}\right)$
- (B) $\cos^{-1}\left(\frac{9}{\sqrt{364}}\right)$
- (C) $\sin^{-1}\left(\frac{5}{2\sqrt{13}}\right)$
- (D) $\sin^{-1}\left(\frac{9}{\sqrt{364}}\right)$

Answer 12: No correct option.

Question 13: The distance of the point (1, 2, 1) from the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{2}$ is

- (A) $\frac{\sqrt{5}}{3}$
- (B) $\frac{2\sqrt{3}}{5}$
- (C) $\frac{20}{3}$
- (D) $\frac{2\sqrt{5}}{3}$

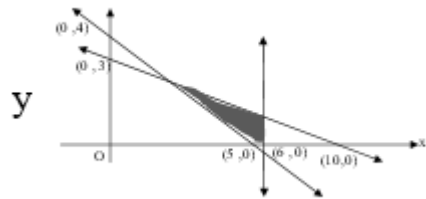
Answer 13: (D)

Question 14: XY – plane divides the line joining the points A (2, 3, -5) and B (-1, -2, -3) in the ratio

- (A) 5 : 3 internally
- (B) 2 : 1 internally
- (C) 5 : 3 externally
- (D) 3 : 2 externally

Answer 14: (C)

Question 15: The shaded region in the figure is the solution set of the in equations.



- (A) $4x + 5y \leq 20, 3x + 10y \leq 30, x \leq 6, x, y \geq 0$
- (B) $4x + 5y \geq 20, 3x + 10y \leq 30, x \leq 6, x, y \geq 0$
- (C) $4x + 5y \leq 20, 3x + 10y \leq 30, x \geq 6, x, y \geq 0$
- (D) $4x + 5y \geq 20, 3x + 10y \leq 30, x \geq 6, x, y \geq 0$

Answer 15: (B)

Question 16: The constant term in the expansion of
$$\begin{bmatrix} 3x+1 & 2x-1 & x+2 \\ 5x+1 & 3x+2 & x+1 \\ 7x-2 & 3x+1 & 4x-1 \end{bmatrix}$$

- (A) -10
- (B) 0
- (C) 6
- (D) 2

Answer 16: (C)

Question 17: If $[x]$ represents the greatest integer function and $f(x) = x - [x] - \cos x$ then

$$f'\left(\frac{\pi}{2}\right) =$$

- (A) 2
- (B) 0
- (C) Does not exist
- (D) 1

Answer 17: (A)

Question 18: If $f(x) = \begin{cases} \frac{\sin 3x}{e^{2x}-1}; x \neq 0 \\ k-2; x = 0 \end{cases}$ is

Continuous at $x = 0$, then $k =$

- (A) $\frac{1}{2}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) $\frac{9}{5}$

Answer 18: No correct options

Question 19: If $f(x) = \sin^{-1} \left[\frac{2^{x+1}}{1+4^x} \right]$, then $f'(x) = 0$

(A) $\frac{2 \log 2}{5}$

(B) $2 \log 2$

(C) $\frac{4 \log 2}{5}$

(D) $\log 2$

Answer 19: (D)

Question 20: If $x = \sec^2 \theta$, $y = a \tan^2 \theta$ then $\frac{d^2 y}{dx^2} =$

(A) 0

(B) $2a$

(C) 4

(D) 1

Answer 20: (A)

Question 21: If α and β are roots of the equation $x^2 - x + 1 = 0$ then $\alpha^2 + \beta^2$ is

(A) $\frac{-1-i\sqrt{3}}{2}$

(B) 1

(C) -1

(D) $\frac{-1+i\sqrt{3}}{2}$

Answer 21: (C)

Question 22: The number of 4 digit numbers without repetition that can be formed using the digits 1, 2, 3, 4, 5, 6, 7 in which each number has two odd digits and two even digits is

(A) 450

(B) 432

(C) 454

(D) 436

Answer 22: (B)

Question 23: The number of terms in the expansion of $(x^2 + y^2)^{25} - (x^2 - y^2)^{25}$ after simplification is

(A) 26

(B) 0

(C) 50

(D) 13

Answer 23: (D)

Question 24: The third term of a G.P. is 9. The product of its first five terms is

(A) 3^{10}

(B) 3^5

(C) 3^{12}

(D) 3^9

Answer 24: (A)

Question 25: A line cuts off equal intercepts on the co-ordinate axes. The angle made by this line with the positive direction of X-axis is

(A) 120°

(B) 45°

(C) 135°

(D) 90°

Answer 25: (C)

Question 26: The eccentricity of the ellipse $9x^2 + 25y^2 = 225$ is

(A) $\frac{3}{4}$

(B) $\frac{4}{5}$

(C) $\frac{9}{16}$

(D) $\frac{3}{5}$

Answer 26: (B)

Question 27: $\sum_{r=1}^n (2r-1) = x$ then $\lim_{n \rightarrow \infty} \left[\frac{1^3}{x^2} + \frac{2^3}{x^2} + \frac{3^3}{x^2} + \dots + \frac{n^3}{x^2} \right] =$

(A) 1

- (B) $\frac{1}{2}$
 (C) 4
 (D) $\frac{1}{4}$

Answer 27: (D)

Question 28: The negative of the statement “All continuous functions are differentiable.”

- (A) Some continuous functions are not differentiable
 (B) All continuous functions are not differentiable.
 (C) All differentiable functions are continuous.
 (D) Some continuous functions are differentiable

Answer 28: (A)

Question 29: Mean and standard deviation of 100 items are 50 and 4 respectively. The sum of all squares of the items is

- (A) 266000
 (B) 251600
 (C) 261600
 (D) 256100

Answer 29: (B)

Question 30: Two letters are chosen from the letters of the word ‘EQUATIONS’. The probability that one is vowel and the other is consonant is

- (A) $\frac{3}{9}$
 (B) $\frac{8}{9}$
 (C) $\frac{5}{9}$
 (D) $\frac{4}{9}$

Answer 30: (C)

Question 31: $f : R \rightarrow R$ and $g : [0, \infty) \rightarrow R$ is defined by $(x) = x^2$ and $g(x) = \sqrt{x}$. Which one of the following is not true?

- (A) $\text{fog}(2) = 2$
 (B) $\text{gof}(4) = 4$
 (C) $\text{gof}(-2) = 2$
 (D) $\text{fog}(-4) = 4$

Answer 31: (D)

Question 32: If $A = \{x \mid x \in \mathbb{N}, x \leq 5\}$, $B = \{x \mid x \in \mathbb{Z}, x^2 - 5x + 6 = 0\}$, then the number of onto functions from A to B is

- (A) 30
- (B) 2
- (C) 32
- (D) 23

Answer 32: (A)

Question 33: On the set of positive rational, a binary operation $*$ is defined by $a * b = \frac{2ab}{5}$. If

$2 * x = 3^{-1}$, then $x =$

- (A) $\frac{2}{5}$
- (B) $\frac{1}{6}$
- (C) $\frac{125}{48}$
- (D) $\frac{5}{12}$

Answer 33: (C)

Question 34: $\cos \left[2 \sin^{-1} \frac{3}{4} + \cos^{-1} \frac{3}{4} \right] =$

- (A) $\frac{3}{5}$
- (B) $\frac{-3}{4}$
- (C) does not exist
- (D) $\frac{3}{4}$

Answer 34: (B)

Question 35: If $a + \frac{\pi}{2} < 2 \tan^{-1} x + 3 \cot^{-1} x < b$ then 'a' and 'b' are respectively.

- (A) 0 and 2π
- (B) 0 and π
- (C) $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

(D) $\frac{\pi}{2}$ and 2π

Answer 35: (D)

Question 36: If $|3x - 5| \leq 2$ then

(A) $1 \leq x \leq \frac{9}{3}$

(B) $-1 \leq x \leq \frac{7}{3}$

(C) $-1 \leq x \leq \frac{9}{3}$

(D) $1 \leq x \leq \frac{7}{3}$

Answer 36: (D)

Question 37: A random variable 'X' has the following probability distribution:

X	1	2	3	4	5	6	7
P(X)	k-1	3k	k	3k	$3k^2$	k^2	k^2+k

Then the value of k is

(A) $\frac{2}{7}$

(B) $\frac{1}{5}$

(C) $\frac{1}{10}$

(D) 2

Answer 37: (B)

Question 38: If A and B are two events of a sample space S such that $P(A) = 2.0$, $P(B) = 0.6$ and $P(A|B) = 0.5$ then $P(A'|B) =$

(A) $\frac{1}{3}$

(B) $\frac{3}{10}$

(C) $\frac{1}{3}$

(D) $\frac{2}{3}$

Answer 38: (A)

Question 39: If 'X' has a binomial distribution with parameters $n = 6$, p and $P(X = 2) = 12$, $P(X = 3) = 5$ then $P =$

(A) $\frac{1}{2}$

(B) $\frac{5}{12}$

(C) $\frac{5}{16}$

(D) $\frac{16}{21}$

Answer 39: Wrong question.

Question 40: A man speaks truth 2 out of 3 times. He picks one of the natural numbers in the set $S = \{1, 2, 3, 4, 5, 6, 7\}$ and reports that it is even. The probability that it is actually even is

(A) $\frac{1}{10}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{1}{5}$

Answer 40: (C)

Question 41: The order of the differential equation $y = C_1 e^{C_2 + x} + C_3 e^{C_4 + x}$ is

(A) 3

(B) 1

(C) 4

(D) 2

Answer 41: (B)

Question 42: If $|\vec{a}| = 16$, $|\vec{b}| = 4$, then $\sqrt{|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2} =$

(A) 16

(B) 4

(C) 64

(D) 8

Answer 42: (C)

Question 43: If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{2\pi}{3}$ and the projection of $\vec{a}$ in the direction $\vec{b}$ is -2, the $|\vec{a}|$

(A) 2

(B) 4

(C) 1

(D) 3

Answer 43: (B)

Question 44: A unit vector perpendicular to the plane containing the vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-2\hat{i} + \hat{j} + 3\hat{k}$

(A) $\frac{-\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

(B) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$

(C) $\frac{-\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$

(D) $\frac{-\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

Answer 44 : (A)

Question 45: $[\vec{a} + 2\vec{b} - \vec{c}, \vec{a} - \vec{b}, \vec{a} - \vec{b} - \vec{c}] =$

(A) $2[\vec{a}, \vec{b}, \vec{c}]$

(B) 0

(C)

(D) $[\vec{a}, \vec{b}, \vec{c}]$

Answer 45: (B)

Question 46: $\sqrt[3]{y}\sqrt{x}\sqrt[6]{(x+y)^5}$, then $\frac{dy}{dx} =$

(A) $x - y$

(B) $\frac{x}{y}$

(C) $\frac{y}{x}$

(D) $x + y$

Answer 46: (C)

Question 47: Rolle's Theorem is not applicable in which one of the following cases?

(A) $f(x) = |x|$ in $[-2, 2]$

(B) $f(x) = x^2 - 4x + 5$ in $[1, 3]$

(C) $f(x) = [x]$ in $[2.5, 2.7]$

(D) $f(x) = x^2 - x$ in $[0, 1]$

Answer 47: (A)

Question 48: The interval in which the function $f(x) = x^3 - 6x^2 + 9x + 10$ is increasing in

(A) $[1, 3]$

(B) $(-\infty, 1) \cup (3, \infty)$

(C) $(-\infty, -1) \cup [3, \infty)$

(D) $(-\infty, 1] \cup [3, \infty)$

Answer 48: (D)

Question 49: The sides of an equilateral triangle are increasing at the rate of 4 cm/sec. The rate at which its area is increasing, when the side is 14 cm.

(A) $42 \text{ cm}^2/\text{sec}$

(B) $103 \text{ cm}^2/\text{sec}$

(C) $14 \text{ cm}^2/\text{sec}$

(D) $14\sqrt{3} \text{ cm}^2/\text{sec}$

Answer 49: No correct option.

Question 50: The value of $\sqrt{24.99}$ is

(A) 5.001

(B) 4.999

(C) 4.897

(D) 4.899

Answer 50: Wrong question

Question 51: $\int_{-3}^3 \cot^{-1} dx =$

(A) 6π

(B) 3π

(C) 3

(D) 0

Answer 51: (B)

Question 52: $\int \frac{1}{\sqrt{x} + x\sqrt{x}} dx =$

- (A) $\tan^{-1} \sqrt{x} + C$
- (B) $2 \log(\sqrt{x} + 1) + C$
- (C) $2 \tan^{-1} \sqrt{x} + C$
- (D) $\frac{1}{2} \tan^{-1} \sqrt{x} + C$

Answer 52: (C)

Question 53: $\int \frac{2x-1}{(x-1)(x+2)(x+3)} dx = A \log|x-1| + B \log|x+2| + C \log|x-3| + K$

Then A, B, C are respectively.

- (A) $\frac{1}{6}, \frac{-1}{3}, \frac{1}{3}$
- (B) $\frac{-1}{6}, \frac{1}{3}, \frac{1}{3}$
- (C) $\frac{-1}{6}, \frac{-1}{3}, \frac{1}{3}$
- (D) $\frac{1}{6}, \frac{1}{3}, \frac{1}{5}$

Answer 53: (C)

Question 54: $\int_0^2 [x^2] dx =$

- (A) $5 - \sqrt{2} + \sqrt{3}$
- (B) $5 - \sqrt{2} - \sqrt{3}$
- (C) $-5 - \sqrt{2} - \sqrt{3}$
- (D) $5 + \sqrt{2} - \sqrt{3}$

Answer 54: (B)

Question 55: $\int_0^1 \sqrt{\frac{1+x}{1-x}} dx =$

- (A) $\frac{\pi}{2}$
- (B) $\frac{\pi}{2} - 1$

(C) $\frac{1}{2}$

(D) $\frac{\pi}{2} + 1$

Answer 55: (D)

Question 56: If U is the universal set with 100 element; A and B are two set such that $n(A) = 50$, $n(B) = 60$, $n(A \cap B) = 20$ then $n(A' \cap B') =$

(A) 90

(B) 40

(C) 10

(D) 20

Answer 56: (C)

Question 57: The domain of the function $f : R \rightarrow R$ defined by $f(x) = \sqrt{x^2 - 7x + 12}$ is

(A) $(-\infty, 3] \cap [4, \infty)$

(B) $(-\infty, 3] \cup [4, \infty)$

(C) $(3, 4)$

(D) $(-\infty, 3] \cup (4, \infty)$

Answer 57: (B)

Question 58: If $\cos x = |\sin x|$ the, the general solution is

(A) $x = n\pi + (-1)^n \frac{\pi}{4}, n \in Z$

(B) $x = n\pi \pm \frac{\pi}{4}, n \in Z$

(C) $x = (2n-1)\pi \pm \frac{\pi}{4}, n \in Z$

(D) $x = 2n\pi \pm \frac{\pi}{4}, n \in Z$

Answer 58: (B)

Question 59: $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ =$

(A) 4

(B) 2

(C) 1

(D) 3

Answer 59: (A)

Question 60: If $P(n) : 2^n < n!$ Then the smallest positive integer for which $P(n)$ is true, is

- (A) 4
- (B) 2
- (C) 5
- (D) 3

Answer 60: (A)

